CS & IT ENGINEERING





Digital Logic

DPP -01 Discussion Notes

Boolean Theorems and GATES

By- Chandan Gupta Sir

Question **(MCQ)**

$f = (A + B)\,(A + C)\,(A + \bar{C})$ is equivalent to $\longrightarrow$ $(A + C\cdot\bar{C})\,(\bar{A} + B)$

$A\cdot\bar{A} + A\cdot B = AB$

$A + (B\cdot C\cdot\bar{C}) = A$

$(A+B)\cdot(A+C)\cdot(A+\bar{C})$

A $A + BC$

B $A + B\bar{C}$

C 0

D A

Question **(MCQ)**

A logic function is given as:

$f(A, B, C) = B\bar{C}\left[A + B\bar{C}D + \bar{B}CD + \bar{A}B\bar{C} + \bar{A}\bar{B}\bar{C}\right]$

is equivalent to

$(A+\bar{A}).(A+B\bar{C})$

$(A + B\bar{C} + B\bar{C}D + \bar{B}CD + \bar{A}\bar{B}\bar{C})$

$B\bar{C}(A + B\bar{C} + \bar{B}CD + \bar{A}\bar{B}\bar{C})$

$= B\bar{C}$

A $A\bar{B}CD$

B $B\bar{C}$

C $A\bar{B} + B\bar{C} + CD$

D $AB\bar{C}D$

Question **(NAT)**

If we have 4-variables in a logic function, the number of non-dual logical functions possible are_____.

$n = 4$ - variables $\Longrightarrow N = 2^4 = 16$ terms $\longrightarrow (2^{16}$ → boolean function

$N = 2^n$

$2^{2^3} \longrightarrow 2^8$ → dual boolean functin

$\longrightarrow M = 2^{16} - 2^8 = 2^8(2^8 - 1)$

$= 256 \times 255$

Question **(MCQ)**

A logic function
$f(A, B, C) = (A + B)(\bar{B} + C)(A + C)$, then $\bar{f}$ will be equal to

A $AB + \bar{B}C$

B $\bar{A}\bar{B} + B\bar{C}$

C $\bar{A}\bar{B} + \bar{A}\bar{C}$

D $AB + AC$

Question **(MCQ)**

Which of the following statement is true?

A Dual function f^D is always equals to f. ✗

B NAND is self dual in nature. ✗

C NOT is self dual in nature. ✓

$\overline{A \cdot B} \xleftrightarrow{\text{dual}} \overline{A + B}$ (NAND ↔ NOR)

$\overline{A} \xleftrightarrow{\text{Dual}} \overline{A}$

D Number of self dual function with 3-variables is 8.

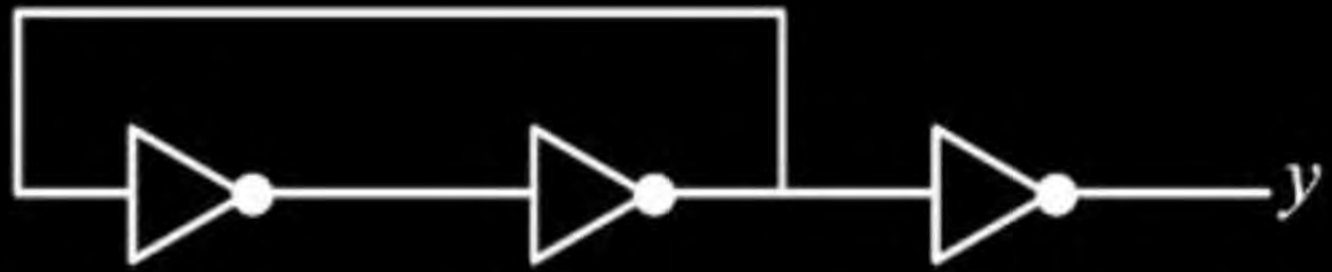
$2^{2^2} = 2^4 = 16 \rightarrow$ self dual function

$2^{2^3} = 2^8 = 256 \rightarrow$ total boolean function

Question **(MCQ)**

A digital circuit is designed as shown:

This circuit works as



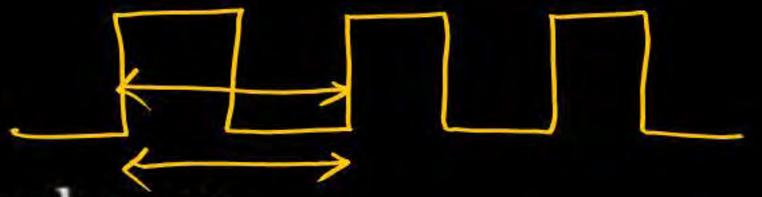
A Astable multivibrator as odd number of NOT gates are used.

B Bistable multivibrator

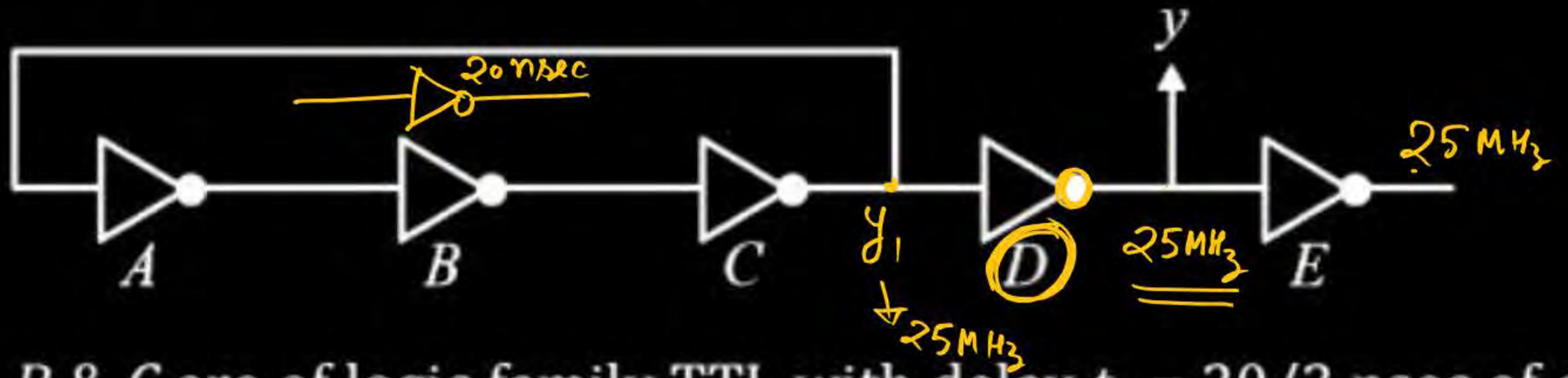
C Monostable multivibrator

D None of these

Question **(NAT)**



An astable multivibrator circuit is designed as shown:



NOT gates A, B & C are of logic family TTL with delay t_d = 20/3 nsec of each gate and gates D & E are of logic family ECL with delay t'_d =10/3 nsec of each gate, then frequency of the waveform at output y is 25 MHz.

$T_{ON} = 20\,nsec$

$T_{off} = 20\,nsec \rightarrow T = 40\,nsec \rightarrow f = \frac{1}{40 \times 10^{-9}} = \frac{10^9}{40}\,Hz = 25\,MHz$

Question **(MCQ)**

Logical function $f(A, B, C, D) = AB + \bar{A}CD + \bar{B}CD$ is equivalent to

A $AB + \bar{B}C$

B $AB + CD$

C $\bar{A}C + \bar{B}C$

D $AB + B\bar{C}$

Question **(MCQ)**

A logical function is given as:
$f(A, B, C) = \bar{A}\bar{B} + \bar{A}BC + \bar{A}B\bar{C}$
then which of the following statement is true?

$f(A,B,C) = \bar{A}\bar{B} + \bar{A}BC + \bar{A}B\bar{C} = \bar{A}\left[\bar{B} + BC + B\bar{C}\right]$

$\bar{A}\bar{B} + \bar{A}B$

$= \bar{A}(\bar{B} + B)$

$\bar{A}\left[\bar{B} + B\right]$

$= \bar{A}$

$f(A,B,C) = \bar{A}$ ✓✓

$f'(A,B,C) = \bar{A}$

A $f(A, B, C) = \bar{A}\bar{B} + B\bar{C}$ ✗

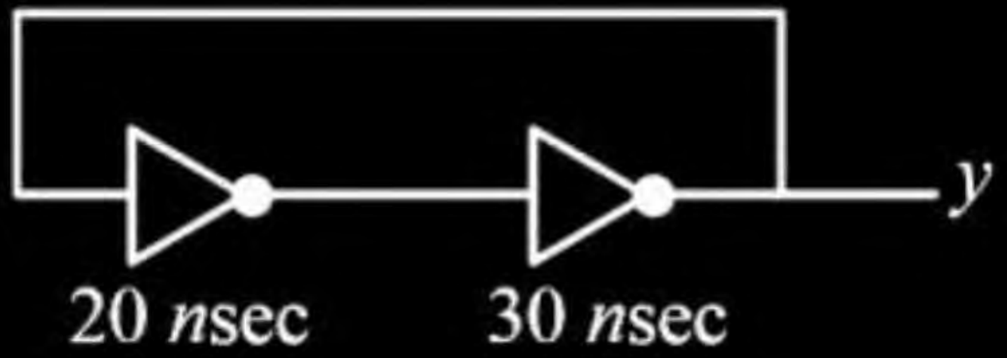
B $f(A, B, C) = \bar{A} + \bar{C}$ ✗

C $f(A, B, C)$ is a self dual function ✓✓

D None of the above

Question **(MCQ)**

A logical circuit is designed as shown:



Output y is

A A bistable multivibrator's output

B A square waveform with $f = 10$ MHz

C A square waveform with $f = 25$ MHz

D None of the above.

Question **(MCQ)**

Which of the following is true?

A $\overline{\bar{A}B + A\bar{B}} = (\bar{A} + \bar{B})(A + B)$

B $\overline{ABC\bar{D}} = \bar{A} + \bar{B} + \bar{C} + \bar{D}$

C $\overline{A\bar{B}.C} = (A + \bar{C})(\bar{B} + \bar{C})$

D None of these

Question **(MCQ)**

Which of the following is true?

A We can use '1' as enable input for OR gate

B We can use '0' as enable input for AND gate

C '0' as well as '1' can be used as enable input for XNOR gate

D None of the these

Question **(MCQ)**

Which of the following relation is true?

A $A \oplus \overline{B} = \overline{A} \odot B$

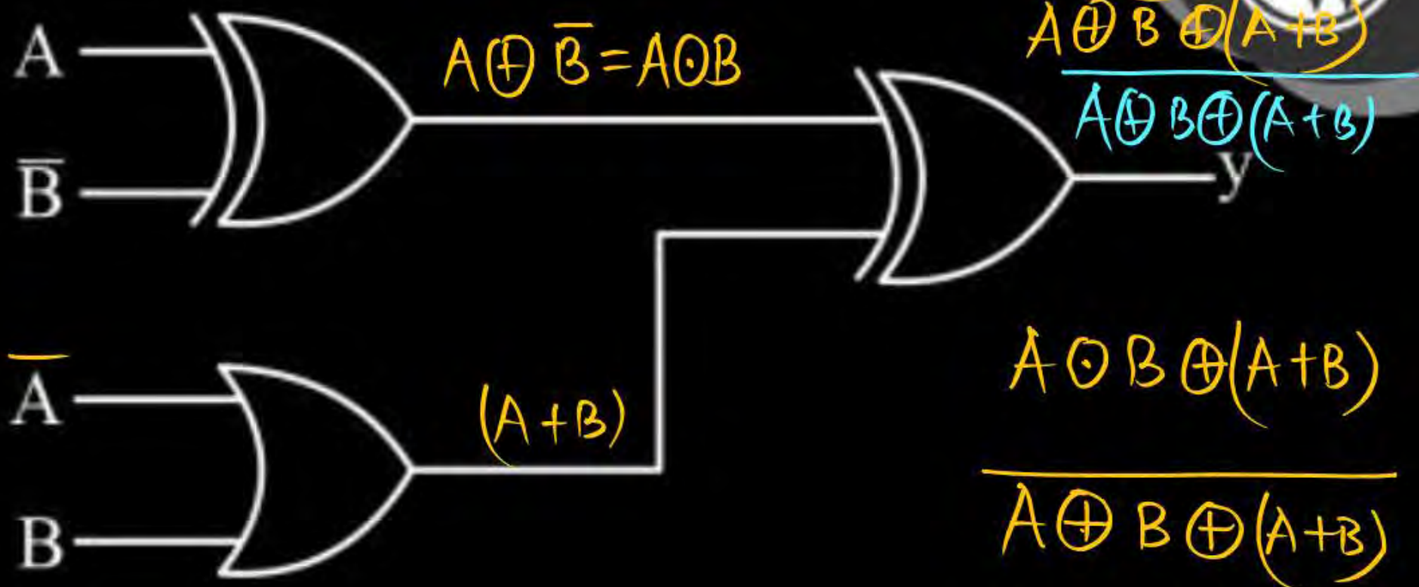
B $\overline{A \oplus \overline{B}} = A \odot B$

C $\overline{\overline{A} \odot \overline{B}} = A \oplus B$

D $\overline{\overline{A} \oplus \overline{B}} = A \oplus B$

Question **(MCQ)**

A logical circuit is as given below:



Output y will be

$A \oplus B \oplus (\bar{A}+B) = A \cdot B$

$A \oplus B \oplus A \cdot B = (A+B)$

$A \oplus \bar{B} = A \odot B$

$\bar{A} \oplus B = A \odot B$

$$\frac{A \oplus \bar{B} \oplus (A+B)}{\bar{A} \oplus B \oplus (A+B)}$$

$$\frac{A \odot B \oplus (A+B)}{\bar{A} \oplus B \oplus (A+B)}$$

$$A \oplus \bar{B} \oplus (\bar{A}+B) = A \oplus B \oplus \overline{(\bar{A}+B)} = \overline{AB} = \bar{A}+\bar{B}$$

$$= \overline{\bar{A} \oplus B \oplus (\bar{A}+B)}$$

$$= \bar{\bar{A}} \oplus B \oplus (\bar{A}+B)$$

$$= \bar{A} \cdot B$$

A $\bar{A}+B$

B $\bar{A}+\bar{B}$

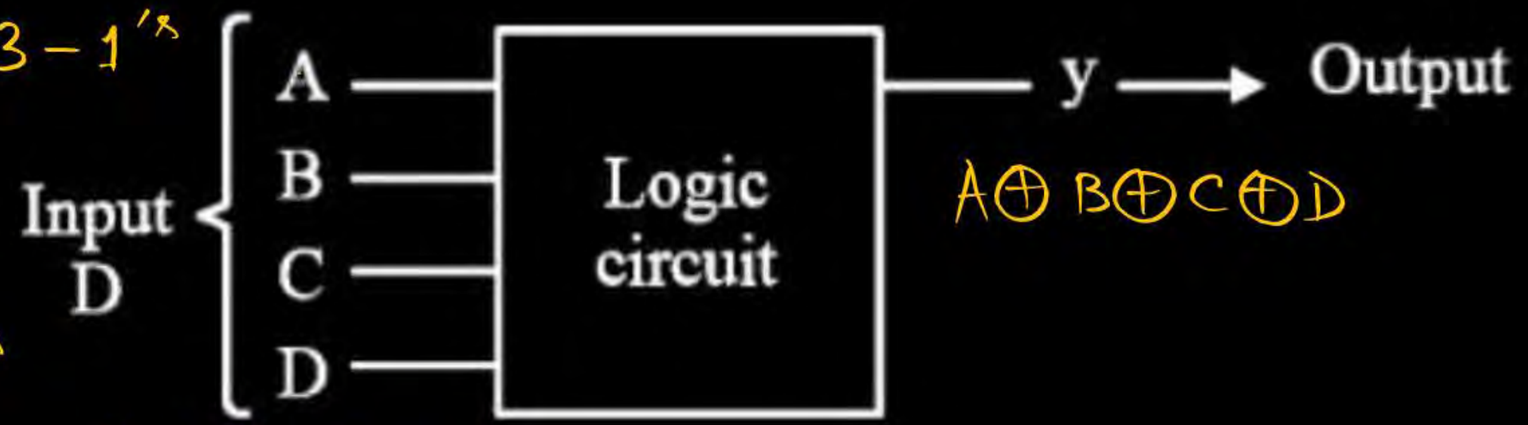
C $A\bar{B}$

D $A+B$

Question **(MSQ)** (B, C)

A logic circuit has 4-input & 1-output line as shown: $A \oplus B \oplus C \oplus D$

Output y is '1' wherever number of zeroes on input side are odd, then output y can be expressed as:



1 zero + 3 − 1's

⟶ 1

3-zeros + 1 − 1's

↓

1

0, 1, 2, 3, 4

↓ ↓

(1, 3)

$\overline{A \odot B \odot C \odot D}$

$= A \oplus B \oplus C \oplus D$

A ~~$A \odot B \odot C \odot D$~~

B $\overline{A \odot B \odot C \odot D}$

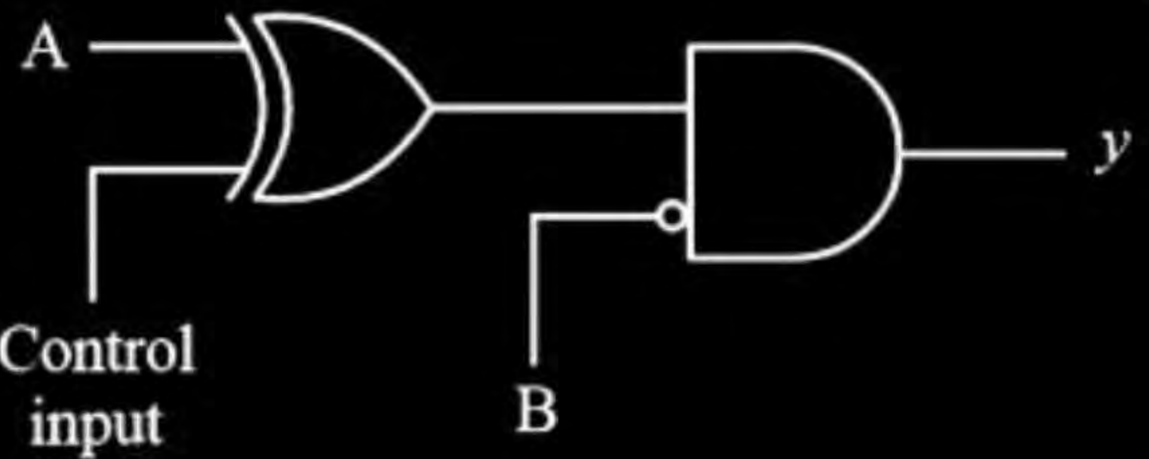
C $\overline{A \oplus B} \oplus C \oplus \overline{D}$

D None of these

$\overline{A \odot B \odot C \odot D}$

Question **(MCQ)**

A logic circuit is as given below:



Which of the following is true?

A Output y is $\bar{A}B$ if control input = 0

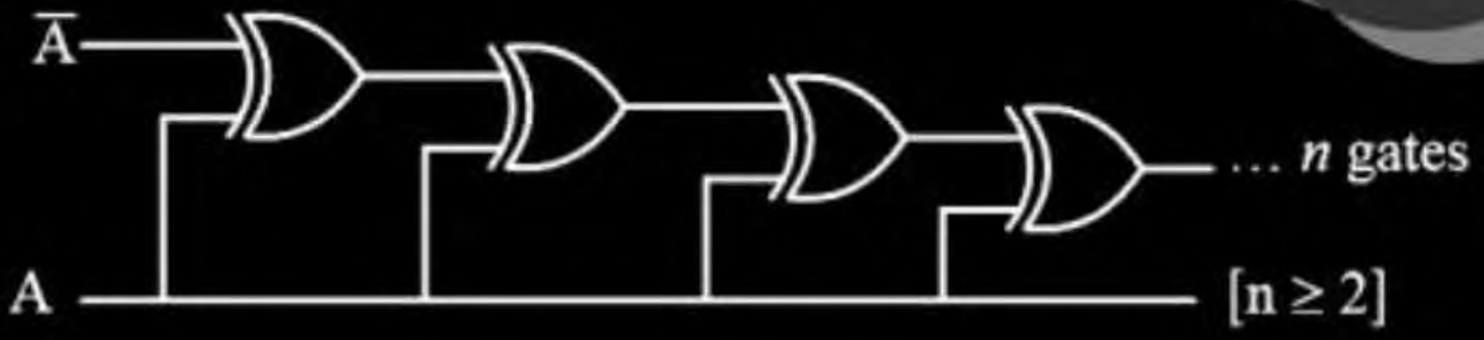
B Output y is $\overline{A+B}$ if control input = 1

C Output y is $\overline{A \cdot B}$ if control input = 0

D Output y is $\overline{\bar{A} \cdot B}$ if control input = 1

Question **(MCQ)**

A logic circuit is as given below:



Which of the following is true?

A Output is $\bar{A}$ if n is even

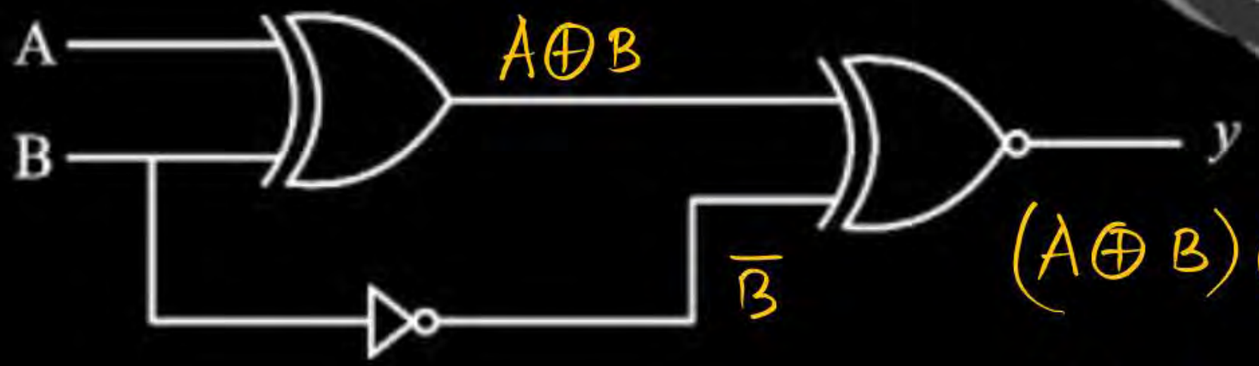
B Output is A if n is even

C Output is $\bar{A}$ if n is odd

D Output is A if n is odd

Question **(MCQ)**

A logical circuit is as given below:



Output y is

A A

B $\bar{B}$

C $\bar{A}$

D B

$(A \oplus B) \odot \bar{B}$

$A \oplus B \odot \bar{B}$

$A \oplus 0$

$= A$

Question **(NAT)**

A logical expression is given as:

$$f(A,B,C,D) = \bar{A} + AB[ABC + \bar{B}C + AB\bar{C} + C\bar{D}]$$

AB

$A\,B \cdot (AB + \bar{B}C + C\bar{D}) = AB$

The minimum number of 2-input NAND gate required to implement above logic function will be 2 .

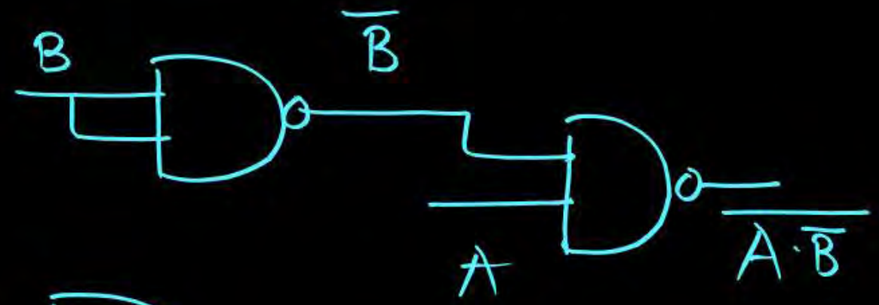
$= \bar{A} + (A \cdot B)$

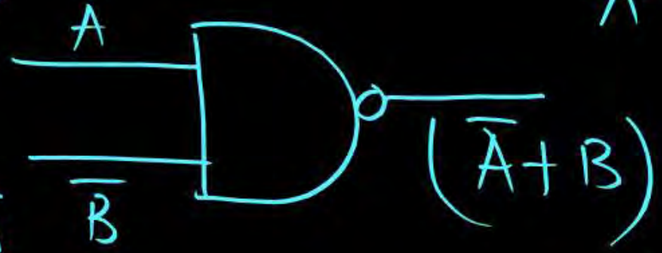
$= (\bar{A} + A)(\bar{A} + B)$

$f = \bar{A} + B$

$\bar{f} = \overline{\bar{A} + B} = A \cdot \bar{B}$

$f = \overline{\underline{A} \cdot \underline{\bar{B}}} = \overline{A \cdot P}$





Question **(NAT)**

A logical expression is given as:

$f(A, B, C) = (\bar{A} + B)(A + \bar{B})$, the minimum number of 2-input NAND gate required to implement above logical function is 5 .

$= A \odot B$

$(\bar{A} + \bar{B})(A + B) = A \oplus B$

Question **(NAT)**

A logical expression is given as:

$f\ (A,\ B,\ C) = \bar{A} + A(BC)$ then minimum number of 2-input NAND gate required to implement above logical function is 2 .

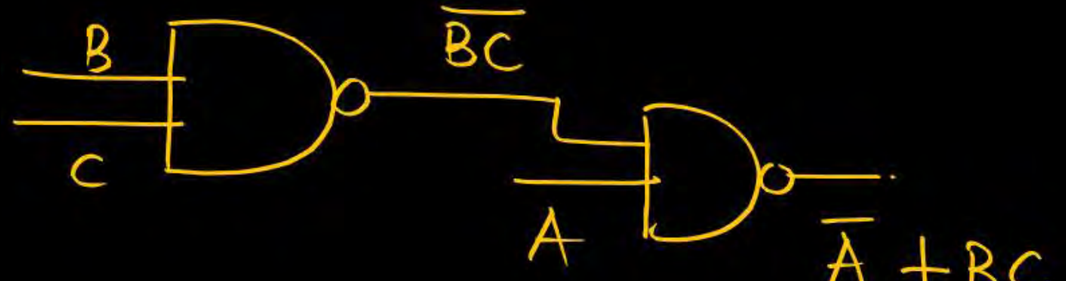
$$= (\bar{A} + A) \cdot (\bar{A} + (BC))$$

$$= (\bar{A} + B)(\bar{A} + C)$$

$$f = \overline{\bar{A} + BC}$$

$$\bar{f} = A \cdot \overline{(BC)} = \overline{A \cdot P} \Rightarrow (\text{NAND})$$

$P = \overline{BC} \rightarrow (\text{NAND})$



Question **(NAT)**

A logical function is given as:

$f(A, B) = A \oplus (A \cdot \overline{B})$ If we implement this logical function using 2-input NAND gate, the minimum number of NAND gate required is 2 .

$$= (A \oplus A) \cdot (A \oplus \overline{B})$$

$$= \overline{A}\,A\overline{B} + A\,\overline{A \cdot \overline{B}}$$

$$= 0 + A(\overline{A} + B)$$

$$= A \cdot B$$

$f_1 = A \cdot \overline{B} \rightarrow 3$

1NAND

2NAND

t.me/chandanguptasirpw

Thank you GW Soldiers !

